A Tale of Five Decoders.

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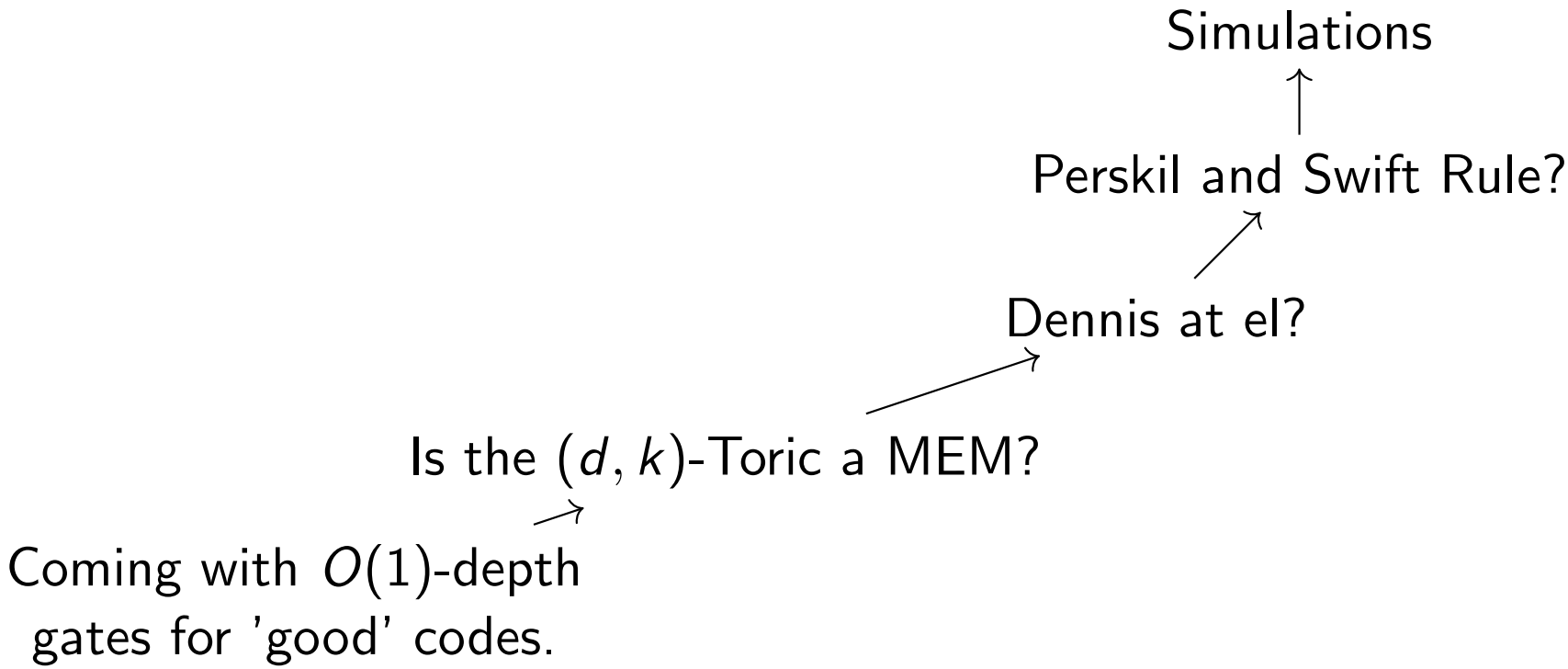
November 3, 2025

Introduction

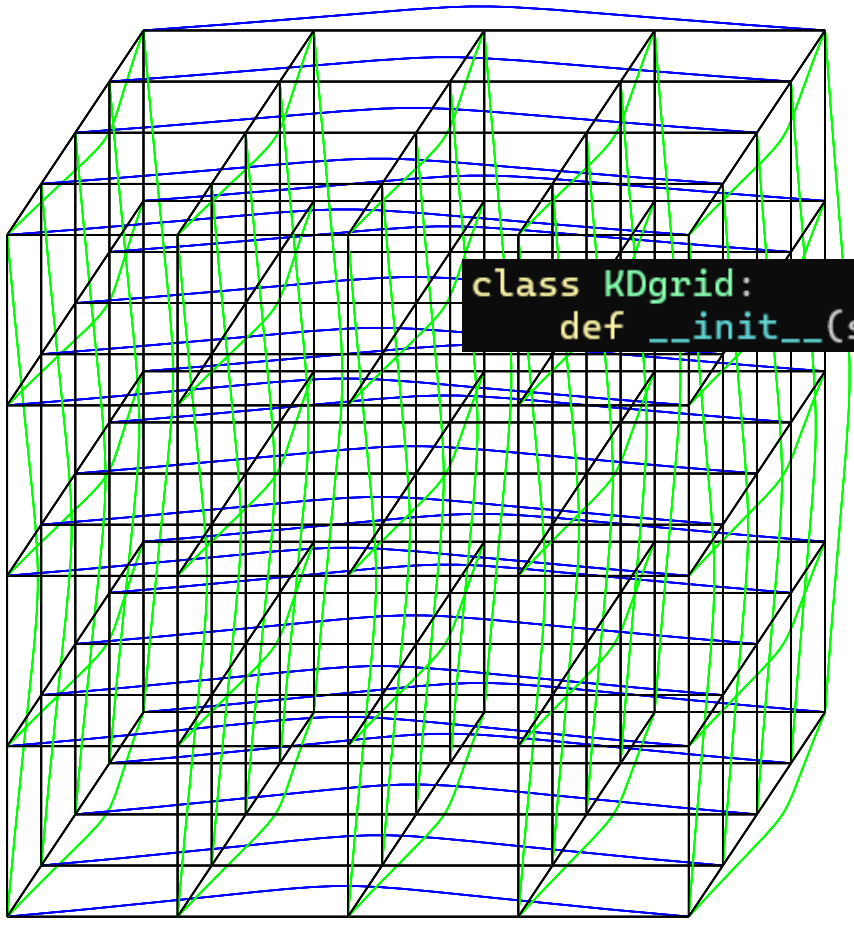
Today:

- Recap, and were we stood before the summer.
- Simulations Results.
- Where we heading.

Map

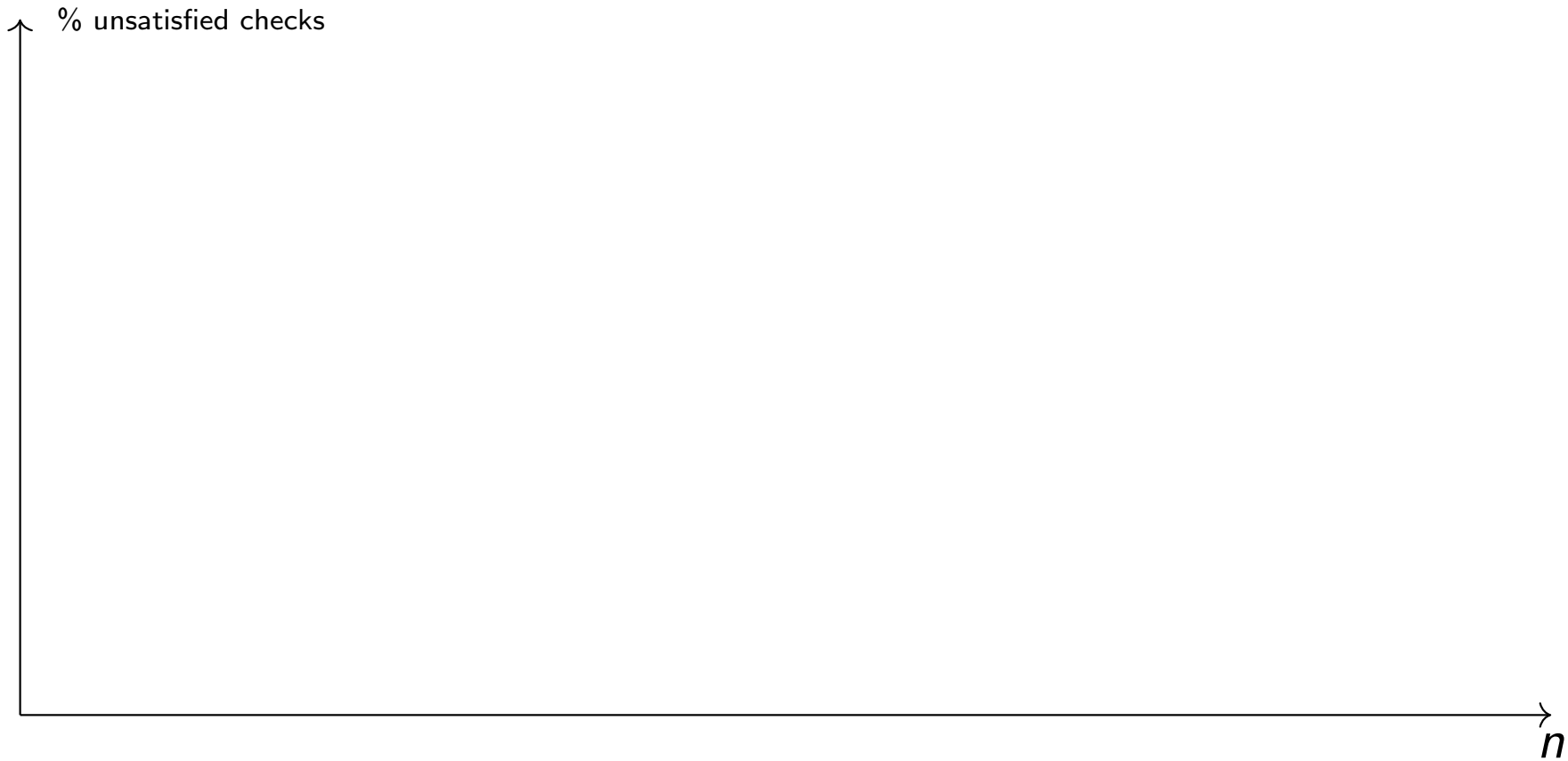

Simulations
Perskil and Swift Rule?
Dennis at el?
Is the (d, k)-Toric a MEM?
Coming with $O(1)$-depth gates for 'good' codes.

The Code



```
class KDgrid:
    def __init__(self, n, k, celldim=2, checksdim =0,
```

How to Read



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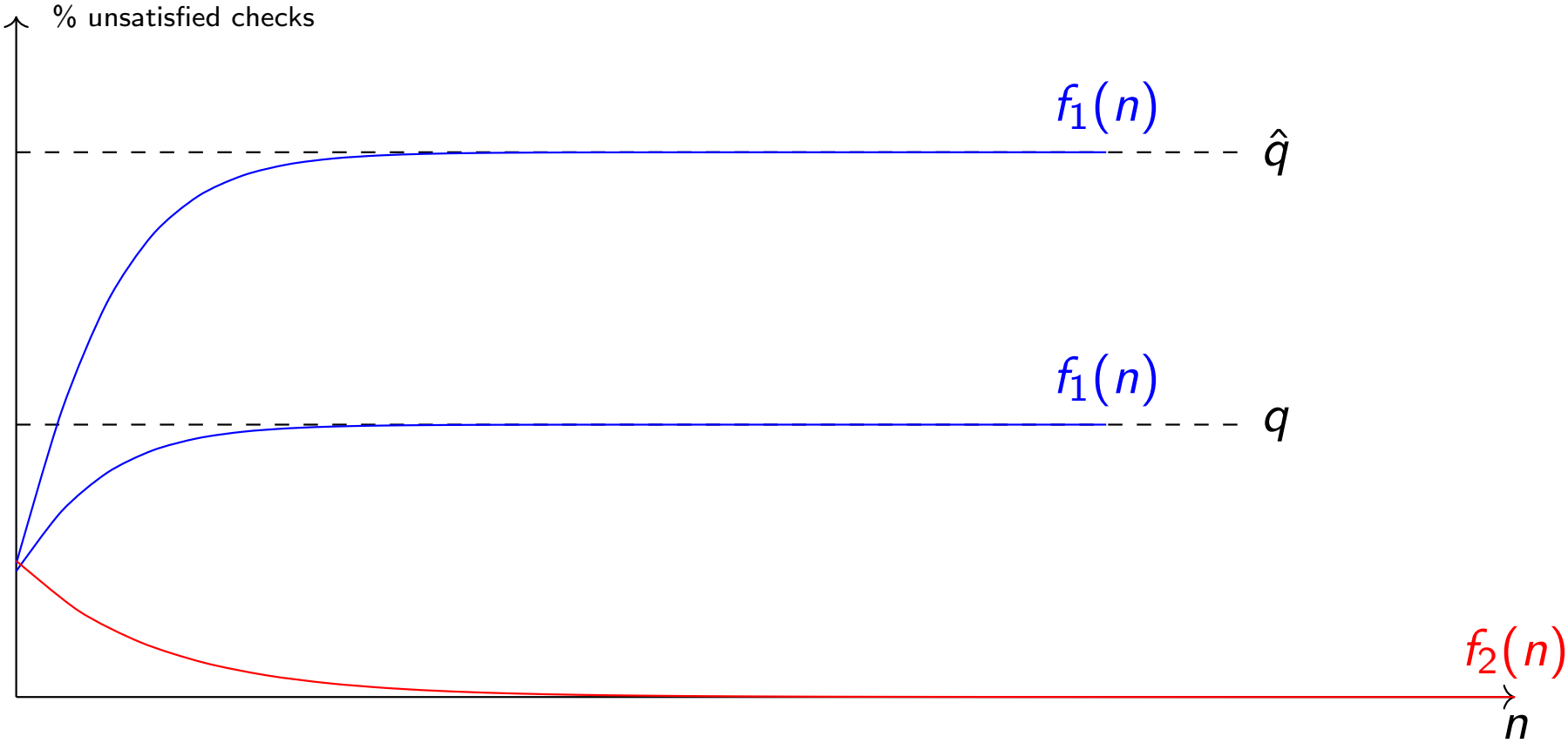
error rates :[0.001]
error accumulation :False

Ideal Result.



colors

In red, No accumulation of errors. In blue decoding in the presence of noise.

Majority

Alg.

Each bit looks at its neighbour checks, if the majority of them is unsatisfied, then it flip itself.

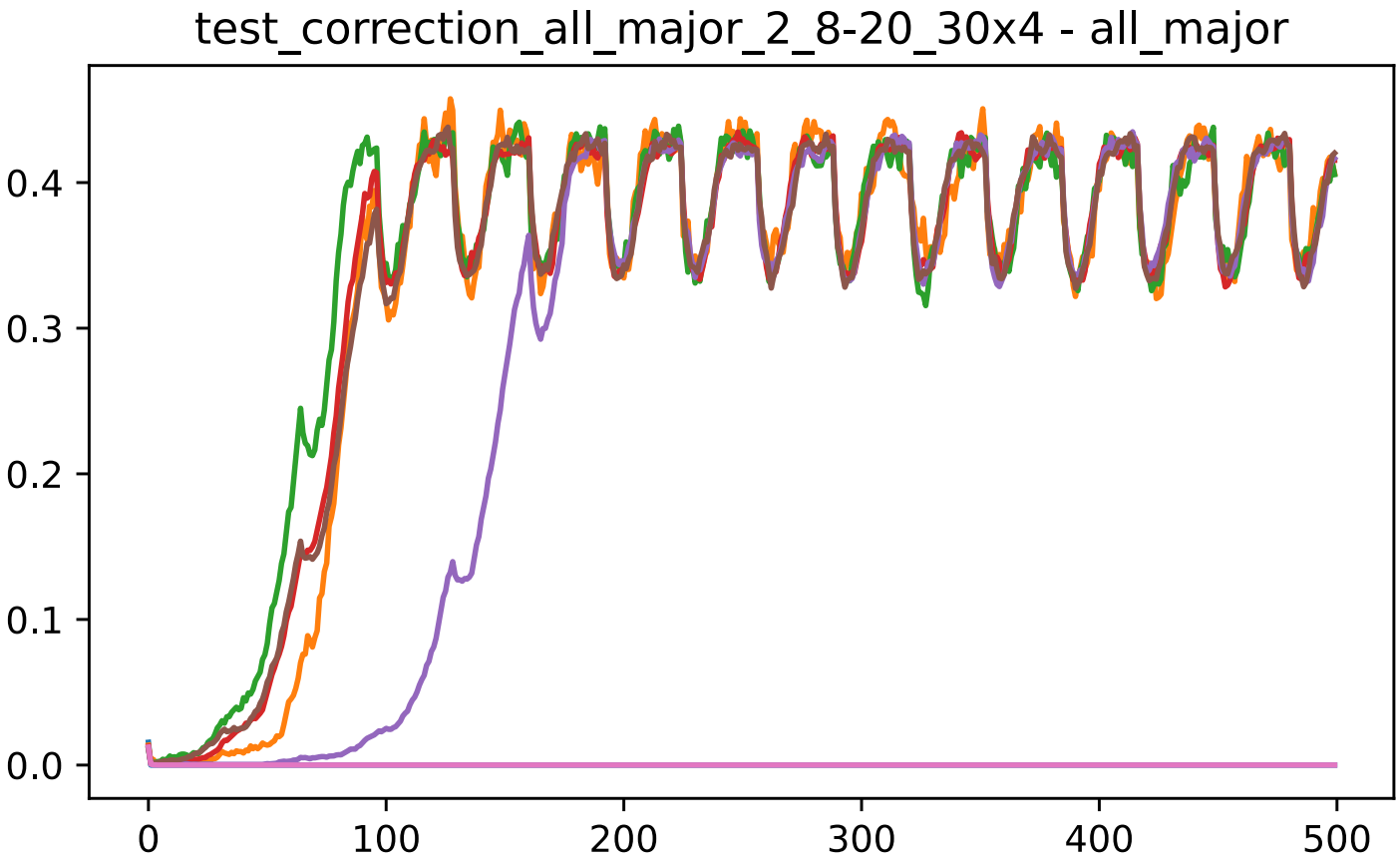
Justification.

Was suggested in Dennis.

Bottom line.

- Fails to correct, even in the absence of noise.
- Collisions.
- Gives better result when requiring $(\frac{1}{2} + \mu)$-majority.

Majority



side-length :[8, 16, 20, 30, 30, 30, 30]
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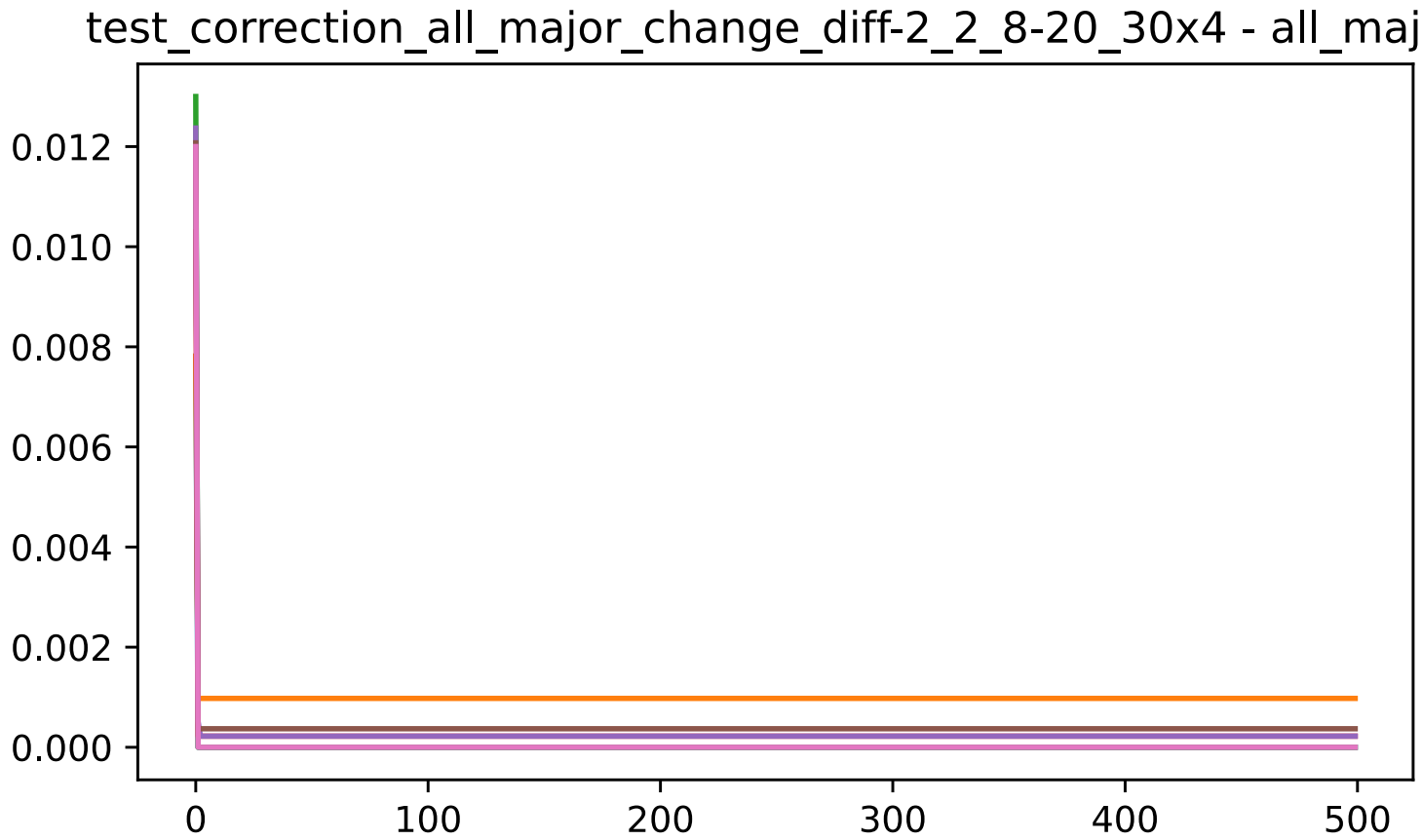
error rates :[0.001]
error accumulation :False

Majority



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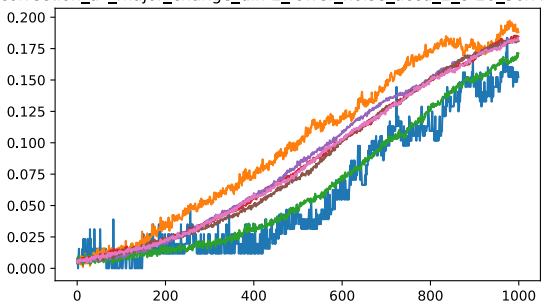
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Majority

test_correction_all_major_change_diff-2_lower_noise_accu_2_8-20_30x4 - all_major



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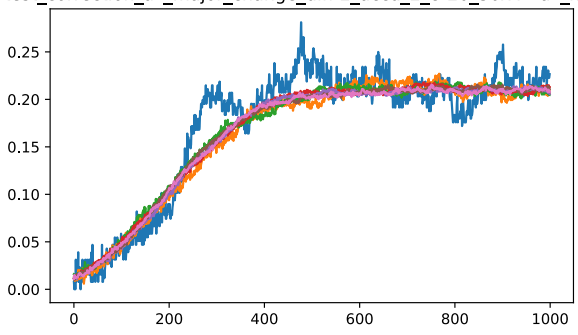
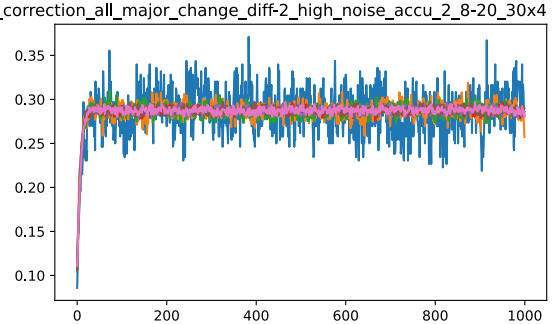
error accumulation :True

test_correction_all_major_change_diff-2_accu_2_8-20_30x4 - all_major



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

test_correction_all_major_change_diff-2_high_noise_accu_2_8-20_30x4 - all_major



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error rates :[0.01]
error accumulation :True

Colors

Alg.

At the initialization, We divide the bits into subsets (colors). Bits of the same color don't share common check. Then, at the correction round ith only bits associated with the color i mod **COLORS** apply a correction.

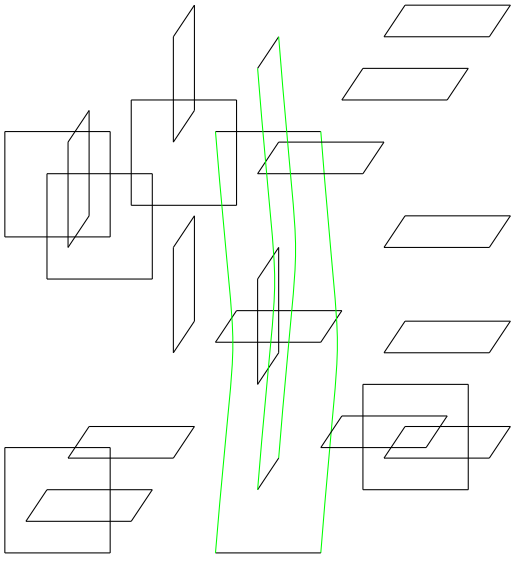
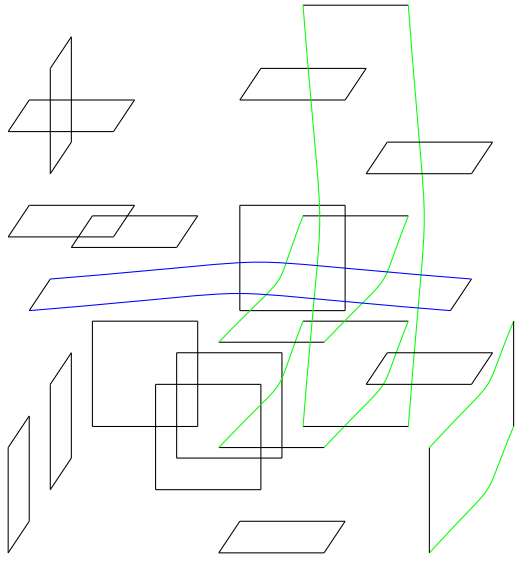
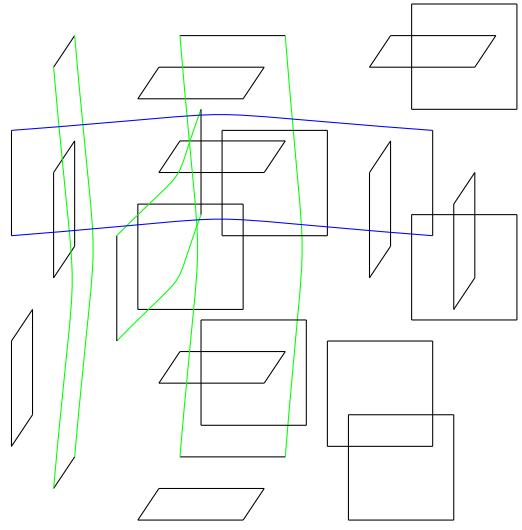
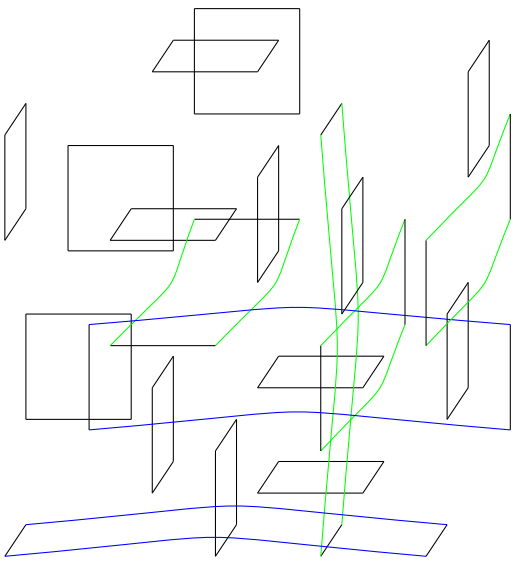
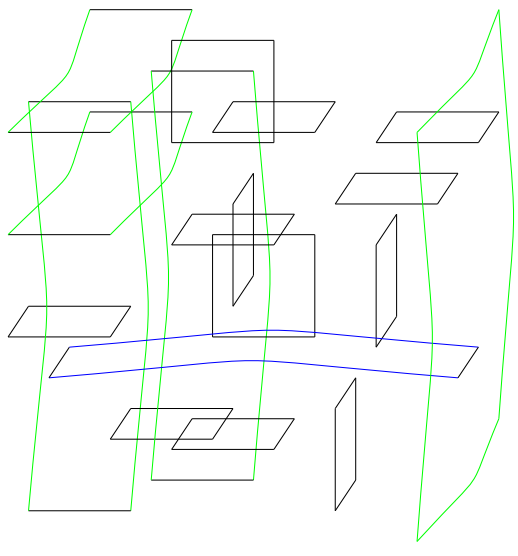
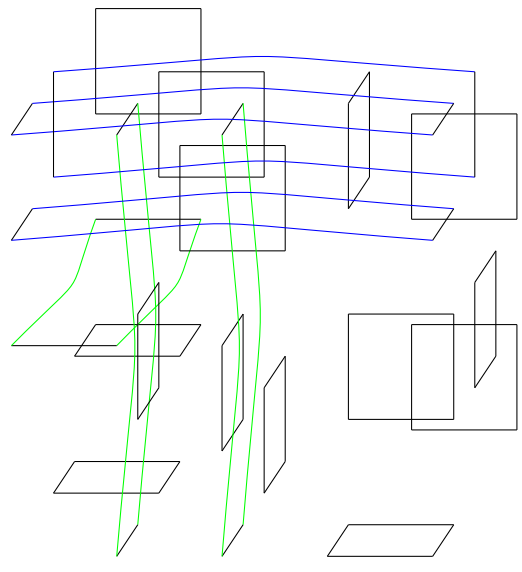
Justification.

1. No collisions, means that bits don't wait until they can read the syndrome.
2. The independence might help when coming to the math.

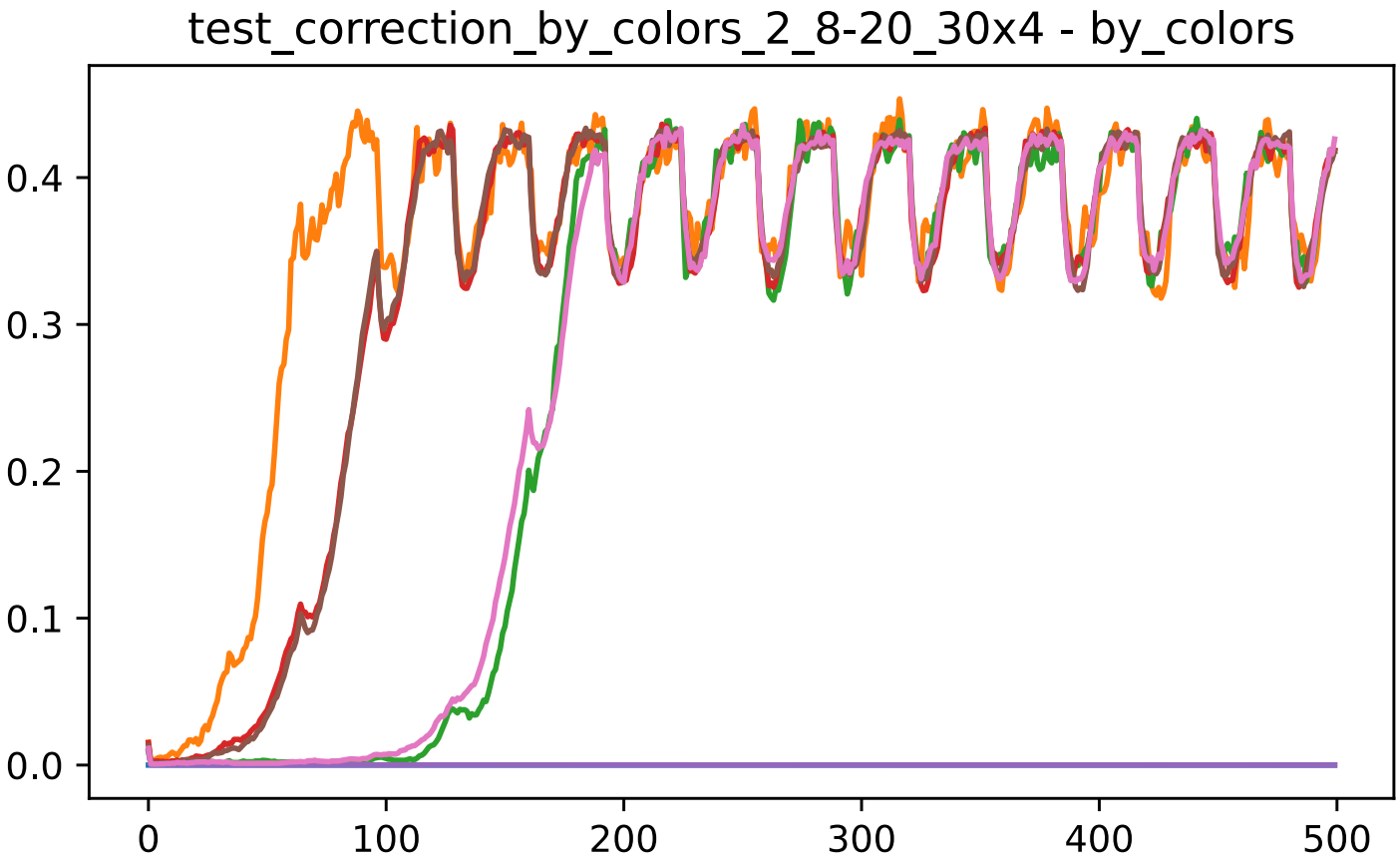
Bottom line.

- I took 32 colors, and sample a random coloring.
- Fails to correct even in the absence of noise.
- Not tested yet when taking $(\frac{1}{2} + \mu)$-majority.

Colors

Colors



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Picking Random Pair

Alg.

Each bit pick two random neighbour checks, and then check if both of them are not satisfied. If so, then flip itself.

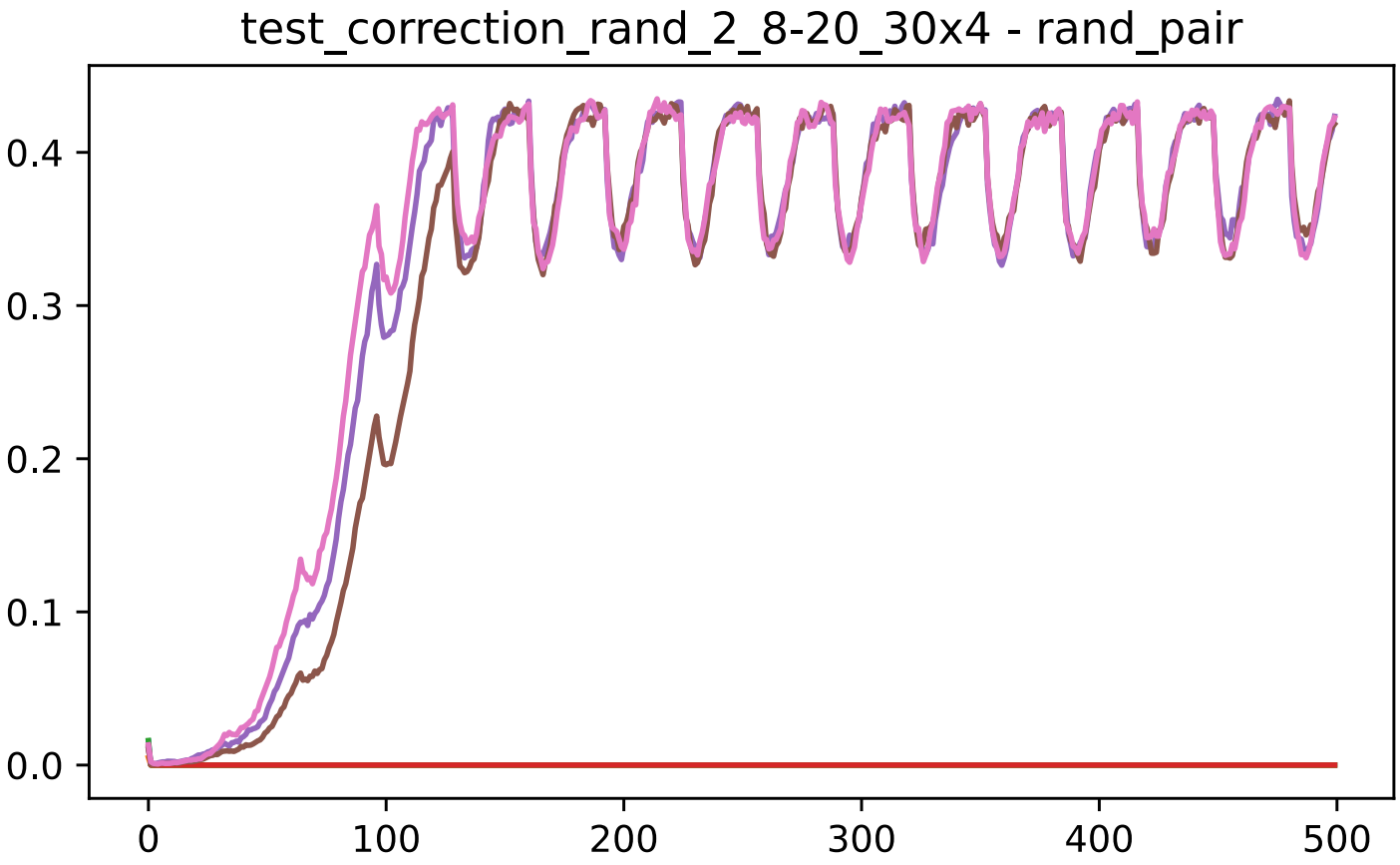
Justification.

With good probability, no too many collisions. Yet, all the bits participate in each round.

Bottom line.

- Fails to correct even in the absence of noise.
- Can we find an analog to $(\frac{1}{2} + \mu)$-majority? Maybe picking 3?

Picking Random



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Swift Rule

Alg.

We define a north vector $\mathbf{1} = (1, 1, 1..1)$, Then each vertex looks only on the cells on its neighbourhood such that they are positive relative to the north vector (Inner product of their (vertices $-$ the reference vertex) with north is positive).
Then, the vertex suggest a correction over those cells, that decreases the local syndrome.

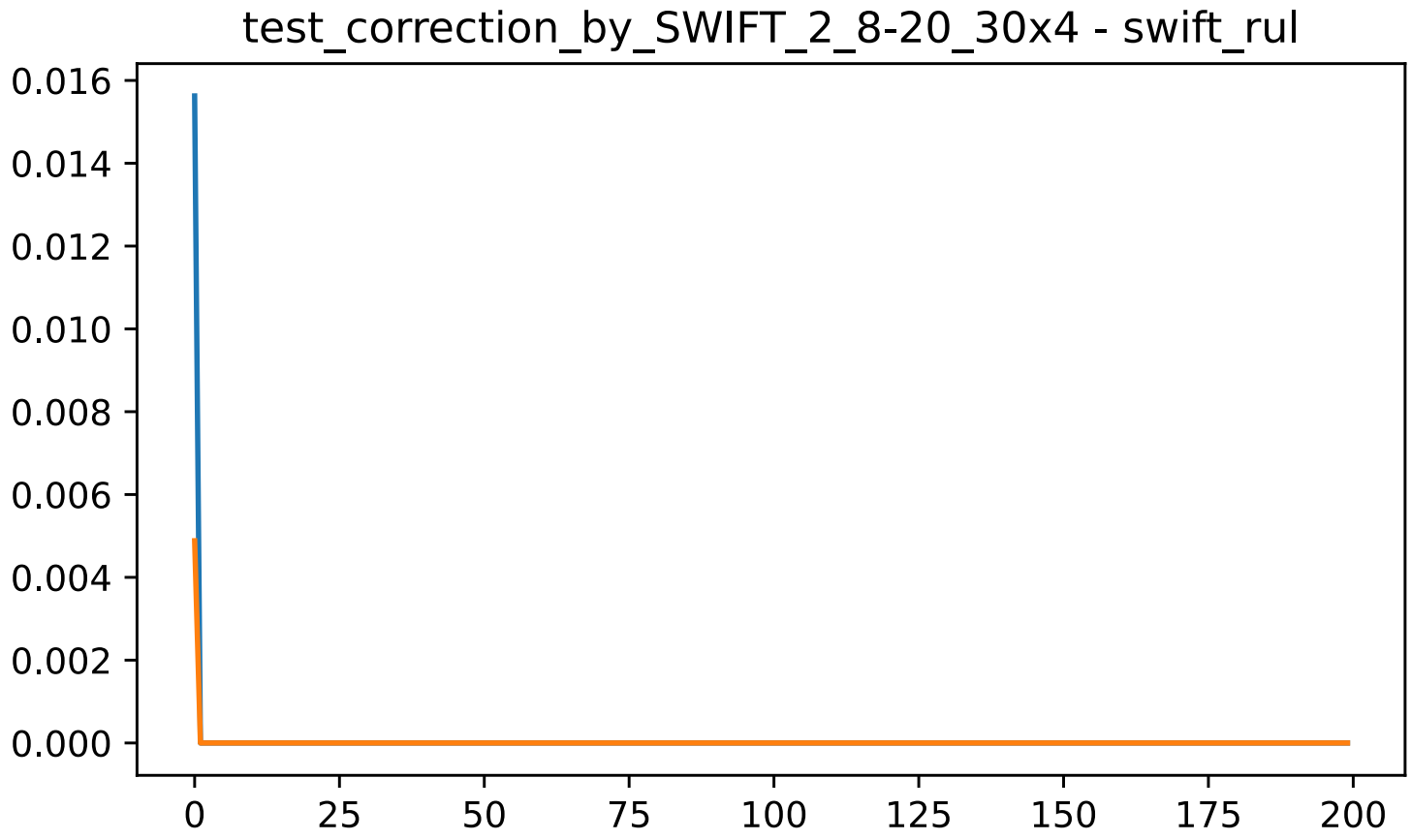
Justification.

1. In the absence of noise, was proven to work.
2. Considered a generalization of Toom's rule.

Bottom line.

- Works in the absence of noise.
- Simulation are too slow for side length of $n > 16$
- Don't have yet results for accumulating setting.

SWIFT



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

SWIFT

```
davidpo+     690  100  1.9 2064736 158696 pts/0  R+   Oct28 4968:07 python ./tests.py
davidpo+     691 99.9  3.0 2152332 245768 pts/0  R+   Oct28 4968:02 python ./tests.py
davidpo+     692 99.9  7.6 2520108 609584 pts/0  R+   Oct28 4967:52 python ./tests.py
davidpo+     741 99.9  7.6 2522152 611480 pts/0  R+   Oct28 4967:42 python ./tests.py
davidpo+     780 99.9  7.7 2526108 615532 pts/0  R+   Oct28 4967:32 python ./tests.py
root         818  0.0  0.0   3064   896 ?        Ss   Oct28   0:00 /init
root         820  0.0  0.0   3080  1024 ?        S    Oct28   0:01 /init
davidpo+     822  0.0  0.0  10480  7116 pts/4    Ss   Oct28   0:00 -zsh
davidpo+     865 99.9  8.1 2564004 653520 pts/0  R+   Oct28 4967:21 python ./tests.py
root        2272  0.0  0.0   3064   896 ?        Ss   Oct28   0:00 /init
```

Max Color

Alg.

We change the setting. We back to the setting that allows a non-nosy computation aside, yet we would like to limit the communication to constant number of bits.

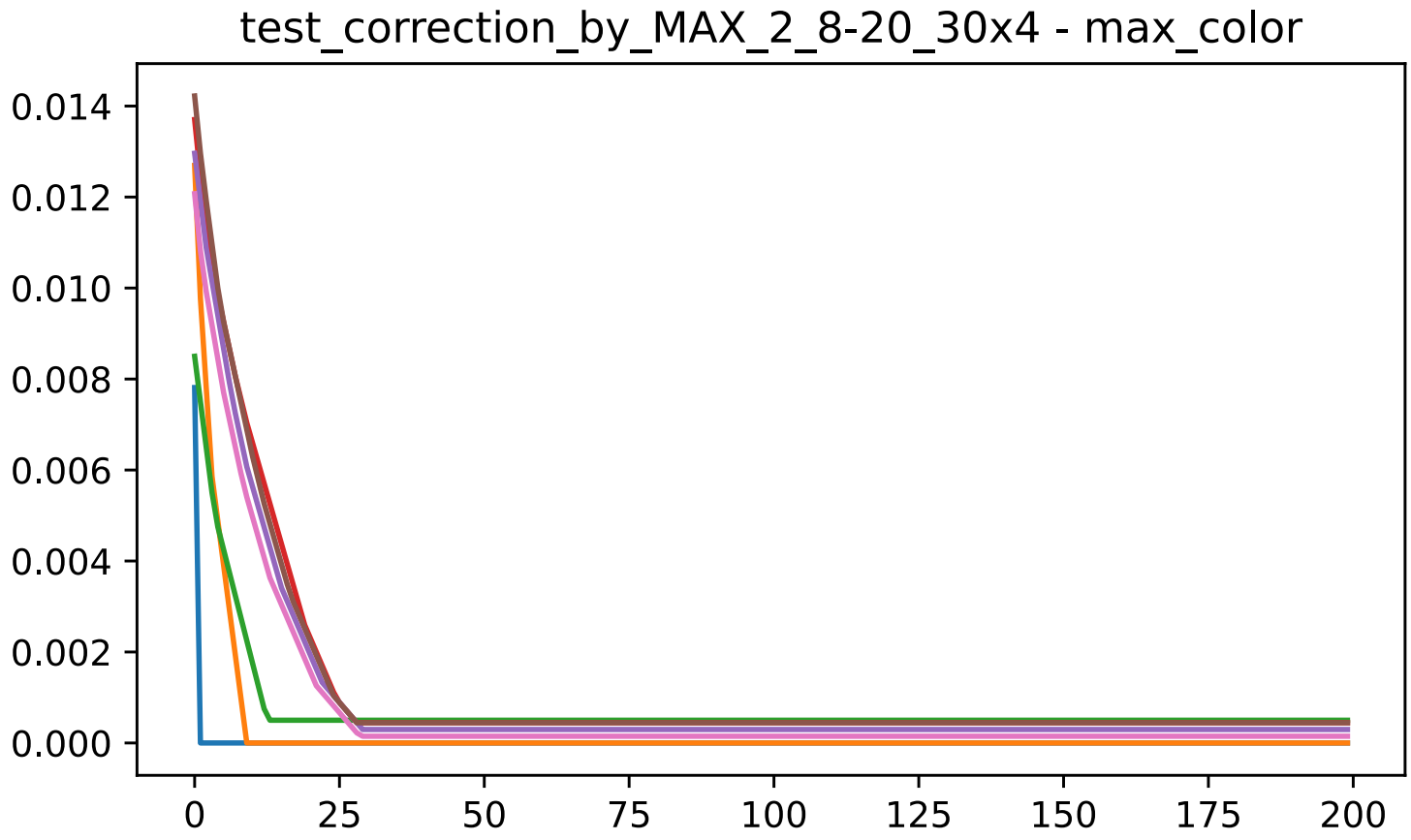
Justification.

1. We don't have works about bounded communication fault tolerance.
2. From mathematical point of view, if the number of colors is c, then we can handle a fraction of $1/c$ of the dirty bits.

Bottom line.

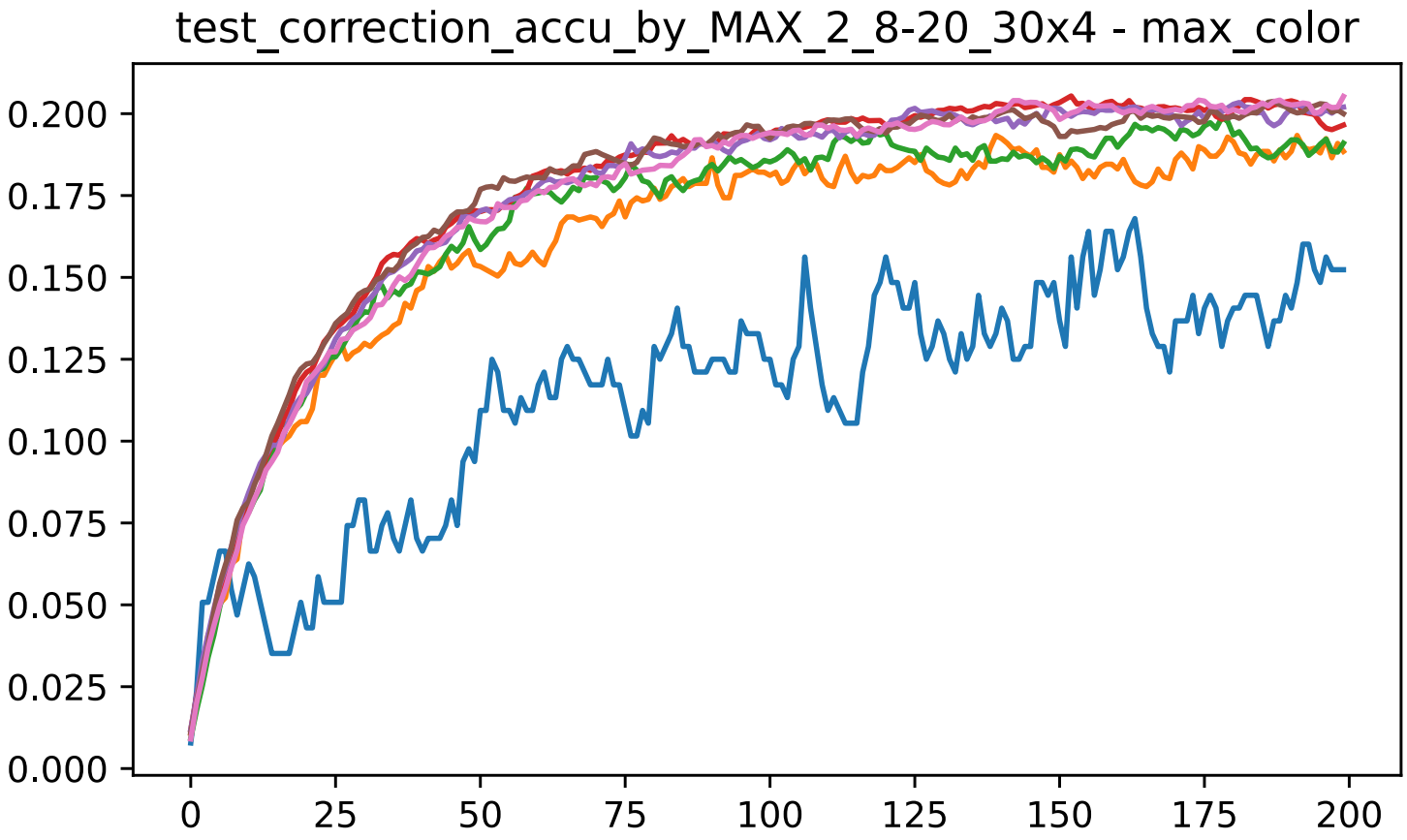
- Works in the absence of noise.
- 'Simulatable'.
- Should be try when vertices correct according to $(\frac{1}{2} + \mu)$ majority.

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True